Experiments of Electrochemical Engineering Course

■ Experiment 1: Solution conductivity measurement

1. Experiment purpose

The purpose of this experiment is to measure the solution conductivity using impedance method and compare the solution conductivity of different solution species, concentration at different temperature.

2. Principle

Introduction to Solution Conductivity

Conductivity (or specific conductance) of an electrolyte solution is a measure of its ability to conduct electricity. The SI unit of conductivity is Siemens per meter (S/m).

Conductivity may be measured by applying an alternating electrical current (I) to two electrodes immersed in a solution and measuring the resulting voltage (V). During this process, the cations migrate to the negative electrode, the anions to the positive electrode and the solution acts as an electrical conductor.

In addition to the concentration of the electrolyte, the electrolyte resistance is directly related to the type and property of the electrolyte (Degree of dissociation, mobility, ionic radius, number of ionic charges, dielectric constant and viscosity of solvent).

In a solution, it obeys the Ohm's Law,

$$I = \frac{U}{R},$$

and the electrolyte resistance can be denoted as follows:

$$R = \frac{\rho l}{a},$$

where ρ is the specific resistance (resistivity), Ω cm.

The Solution conductance (S/cm) is the reciprocal of solution resistivity, which are expressed as follows:

$$\kappa = \frac{1}{\rho}$$

Conductivity measurement methods

The electrical conductivity of a solution of an electrolyte is measured by determining the resistance of the solution between two flat or cylindrical electrodes separated by a fixed distance, as shown in Figure 1.

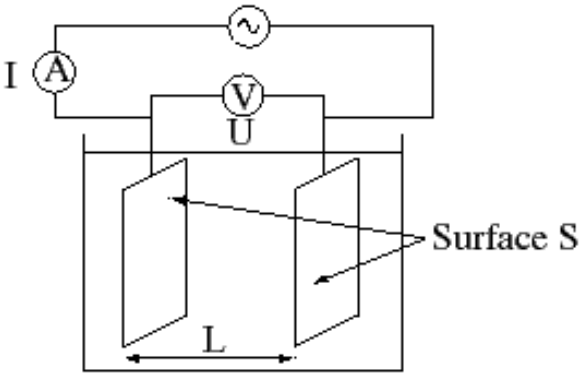


Figure 1 Principle of the measurement

When the electrodes were fabricated, the area of the two electrodes (*S*) and the distance between two electrodes (*L*) are fixed. L/S, therefore, can be used as a constant, electrode constant K, denoted as the following equation,

$$K = \frac{L}{S}.$$

However, the L and S are very difficult to measure. In order to determine the K, we often use a known concentration solution such as standard KCL solution to determine the resistance R, and then determine the K.

3. Experiment apparatus and materials

The experiment apparatus and materials, as shown in Figure 2, contain:

- ✓ Solute: KCl, ZnSO4
- ✓ Solvent: Deionized water
- ✓ Beaker
- ✓ Electronic balance
- ✓ Magnetic mixer
- ✓ Electrodes: Pt sheet
- ✓ Biologic VMP (electrochemical test station)



Figure 2 The apparatus and materials of solution conductivity measurement experiment

4. Experiment plan

Temperature / Concentration	25℃	40℃	55℃	Notes
0.01 mol/L	●			For KCl and ZnSO4 solutions
0.1 mol/L	●	●	●	
0.5 mol/L	●			
1 mol/L	●			
3.4 mol/L	●			Only KCl solution

Figure 3 The plan of solution conductivity measurement experiment

5. Procedure

P1. Get familiar with the experiment principles and the equipment.

P2. Solute weighting: weight corresponding mass of solute using electronic balance.

For KCl solute, the weights are: 12.495 g, 3.675 g, 1.838 g, 0.368 g and 0.037 g;

For $ZnSO_4$ solute, the weights are: 8.05 g, 4.025 g, 0.805 g and 0.081 g.

P3. Solution preparation: put the solute and deionized water (50 mL) into the beaker and mix the solution using magnetic mixer for 30 minutes.

P4. Put two Pt sheet electrodes into the solution and connect the electrodes with electrochemical test station (Biologic VMP).

P5. Open the EC-lab software, setting the following parameters: frequency range, 5000 Hz to 1000 Hz (10 points in total), amplitude, 5 mV.

P6. After setting the parameters, click the green button to run the program and record the resistance of each solution.

P7. Apply a voltage step and record the current response.

P8. Calculate the solution conductivity and answer the questions below in your report.

6. Questions

Analyze the following questions with your experiment results and related literature.

Q1: Calculate the conductivity of the tested solution (two calculation method).

Q2: What is the relationship between solution conductivity and concentration? (Kohlrausch's law)

Q3: How does temperature affect the solution conductivity?

Q4: How does solution species affect the solution conductivity?

Q5: Analyze the factors affecting solution conductivity during your experiment.

*Q*6: Search the literatures related to the solution conductivity test and summarize the common methods to measure conductivity in industry.

Q7: Since the electrodes are composed of two parallel metal plates, in the electrolyte solution, the interface between the solution and the metal electrode has an electric double layer. What be measured actually is the impedance of the tested solution, not the pure resistance, and the equivalent circuit of the tested system is as shown in Figure 4. $R_{\mathrm{L1}}, R_{\mathrm{L2}}$ are the electrode lead resistance of two sides, separately; $C_{\mathrm{DL1}}, C_{\mathrm{DL2}}$ are the electric double layer capacitance of two sides, separately; Z_1, Z_2 are the electrode reaction resistance of two sides, separately; R_{X} is the solution resistance; C_{P} is the electrolyte capacitance.

a) When the conductivity measurement use the AC excitation, and the electrode lead resistance can be neglected, please simply the equivalent circuit and give your reasons.

b) When the conductivity measurement use the AC excitation, the tested frequency is enough high, and the tested solution resistance is high. Please simply the equivalent circuit and give your reasons. In this case, if we give a voltage step to the test system, what's the time constant?

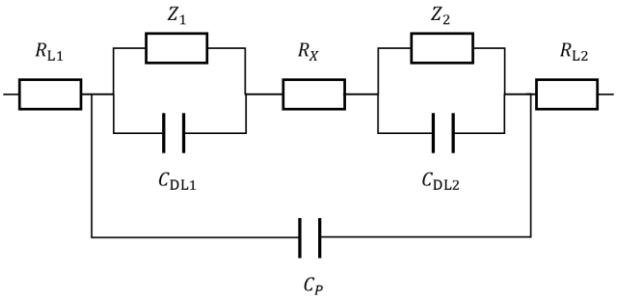


Figure 4 The schematic of equivalent circuit of the test system

■ Experiment 2: Water Electrolysis

1. Experiment purpose:

Electrolyze the water, sulfuric acid solution (H_2SO_4) and potassium chloride (KCl) solution and analyze the corresponding phenomenon.

2. Principle

Faraday's Law,

$$m = \left(\frac{Q}{F}\right)\left(\frac{M}{z}\right)$$

where m is the mass of the substance liberated at an electrode, Q is the total electric charge passed through the substance in coulombs, F = 96485.33289C/mol is the Faraday constant, M is the molar mass of the substance in grams per mol, z is the valence number of ions of the substance (electrons transferred per ion).

Apply the Faraday's Law to analyze the water electrolysis reaction.

$$2H_2O \rightarrow 2H_2 \uparrow + O_2 \uparrow$$

3. Experiment apparatus and material

The experiment apparatus and materials, as shown in Figure 5, contain:

- ✓ KCl and deionized water
- ✓ H_2SO_4 solution
- ✓ Beaker
- ✓ Electronic balance
- ✓ Magnetic mixer
- ✓ Hoffmann electrolyzer
- ✓ DC power supply

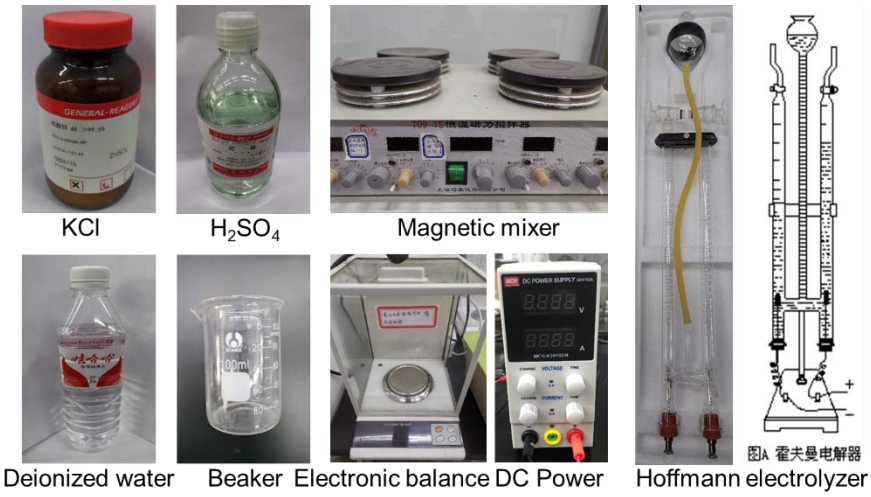


Figure 5 The apparatus and materials of water electrolysis experiment

4. Procedure

P1. Solution preparation: deionized water, 15% H_2SO_4 solution and 1 mol/L KCl solution.

P2. Open the plug, add the solution into the spherical funnel until both sides of tubes are filled up with solution. Then close the plug.

P3. Connect the electrolyzer to DC power supply (setting the voltage of 10 V), there will be bubbles on the two electrodes. Stop after 1 to 2 minutes, and the volume of the cathode-forming gas will be approximately twice the volume of the anode-forming gas. Record the volume of the gas in cathode and anode.

P4. Remove the DC power supply, and connect the two electrodes with LED.

5. Questions

Analyze the following questions with your experiment results and related literature.

Q1. Calculate the total electric charge passed through the electrode during the experiment.

Q2. The volume ratio is not precisely 2:1. Try to explain the reason. (Make comparison between deionized water, H_2SO_4 solution and KCl solution)

Q3. What kinds of electrode should be used in this experiment, and why?

■ Experiment 3: Cyclic voltammetry (CV) measurement

1. Experiment purpose

Measure the cyclic voltammetry (CV) of Pt electrode in H_2SO_4 solution and analyze the effect of solution concentration and scan rate on CV curve

2. Principle or methods

The direct measurement of the potential of a single electrode is not possible without introducing additional interfaces. Only the difference in electrode potential can be measured experimentally. In order to measure the electrode potential of an electrode, we need to measure the difference of potential between the interested one and a reference electrode whose potential is known or predefined. In this experiment, we adopt three-electrode system to measure cyclic voltammetry of Pt in H_2SO_4 solutions, using Pt wire electrode as working electrode, saturated calomel electrode (SCE) as reference electrode and carbon electrode as counter electrode.

3. Experiment apparatus and material

The experiment apparatus and materials, as shown in Figure 6, contain:

- ✓ H_2SO_4 solution
- ✓ Pt electrode
- ✓ SCE
- ✓ Carbon electrode
- ✓ Three electrode bottle
- ✓ Beaker
- ✓ Biologic VMP (electrochemical test station)

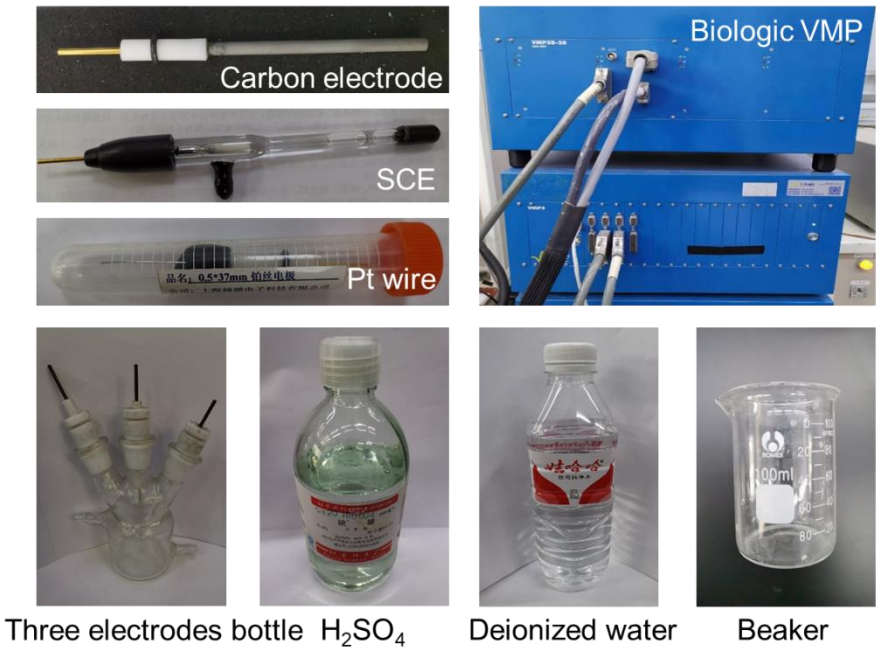


Figure 6 The apparatus and materials of water electrolysis experiment

4. Experiment plan

Concentration / Scan rate	1 mol/L	0.5 mol/L	0.1 mol/L
20 mV/s	●		
40 mV/s	●	●	●
60 mV/s	●		

Figure 7 The plan of cyclic voltammetry measurement experiment

5. Procedure

P1. Solution preparation: three concentrations of H_2SO_4 solution: 1 mol/L, 0.5 mol/L and 0.1 mol/L.

P2. Set up the three-electrode system.

P3. Connect the three electrodes to the electrochemical test station.

P4. Open the EC-lab software, setting the following parameters: voltage range,-1 V to 1 V, scan rate, 20 mV/s (or 40 mV/s, or 60 mV/s).

P5. Analyze the measured CV curve.

6. Questions

Analyze the following questions with your experiment results and related literature.

Q1. Reading literature about the cyclic voltammetry of Pt electrode in H_2SO_4 solution. Introduce the principle in your report.

Q2. Make comparison between different solution concentrations.

Q3. Make comparison between different scan rate.

Experiment work division:

Group No.	Experiment 1	Experiment 2	Experiment 3
A	KCl solution: different concentration	All	Same concentration, different scan rate
B	KCl solution: different temperature	All	Same scan rate, different concentration
C	ZnSO4 solution: different concentration	All	Same scan rate, different concentration
D	ZnSO4 solution: different temperature	All	Same concentration, different scan rate

Note: each group only need to finish the work in the table above during the experiment class, and prepare group presentation based on this. But the report submitted to TAs need to contain all group's work, which means you need to analyze not only your experiment results, but also other group's experiment results. The experiment report should also contain the answer to questions raised in this manual.